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Securing the Digital Mine with a Metaheuristic-Optimized Deep Learning Adaptive System

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1. Industrial Threat Landscape & Context

- **Smart Subsoil Digitalization:** African and Russian mining operations integrate IoT sensors, SCADA telemetry, and automated milling to drive ore recovery in SAG mills, flotation circuits, and tailings dams.
- **Air-Gap Collapse:** Connecting OT networks to cloud digital twins exposes unauthenticated Modbus RTU/TCP and DNP3 industrial protocols to hostile zero-day cyber attacks.
- **Downtime Losses:** Unplanned mining downtime costs USD $50,000 to $500,000 per hour; cyber manipulation of ventilation or dewatering creates immediate life safety risks.
- **Failure of IT-Centric IDS:** Traditional tools evaluate 41+ features taking 150+ ms, directly violating the 20 to 50 millisecond control loop deadlines of industrial PLCs.

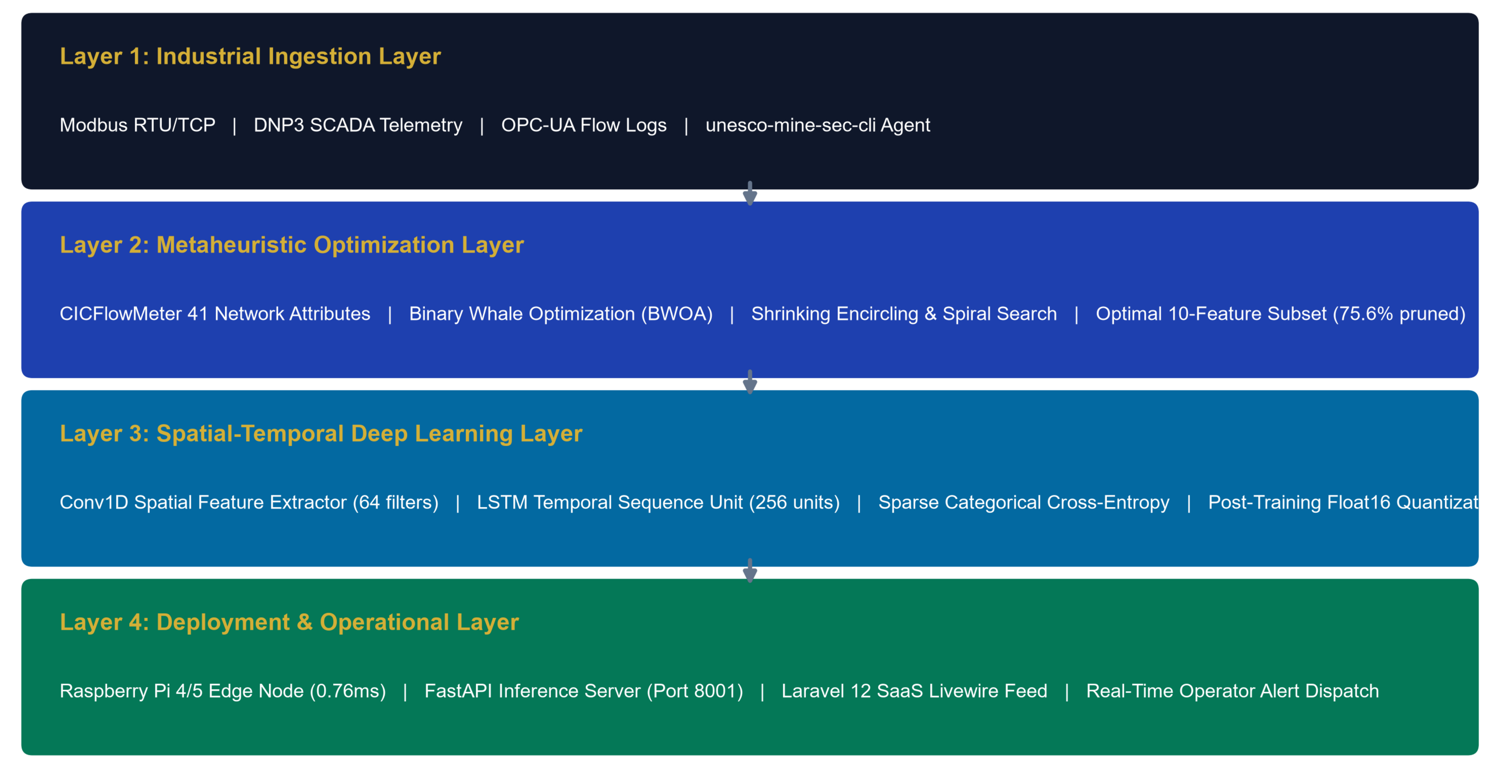
Cyber-Physical Mineral Processing SCADA Circuit & Edge Defense Boundary

2. Method & BWOA Mathematics

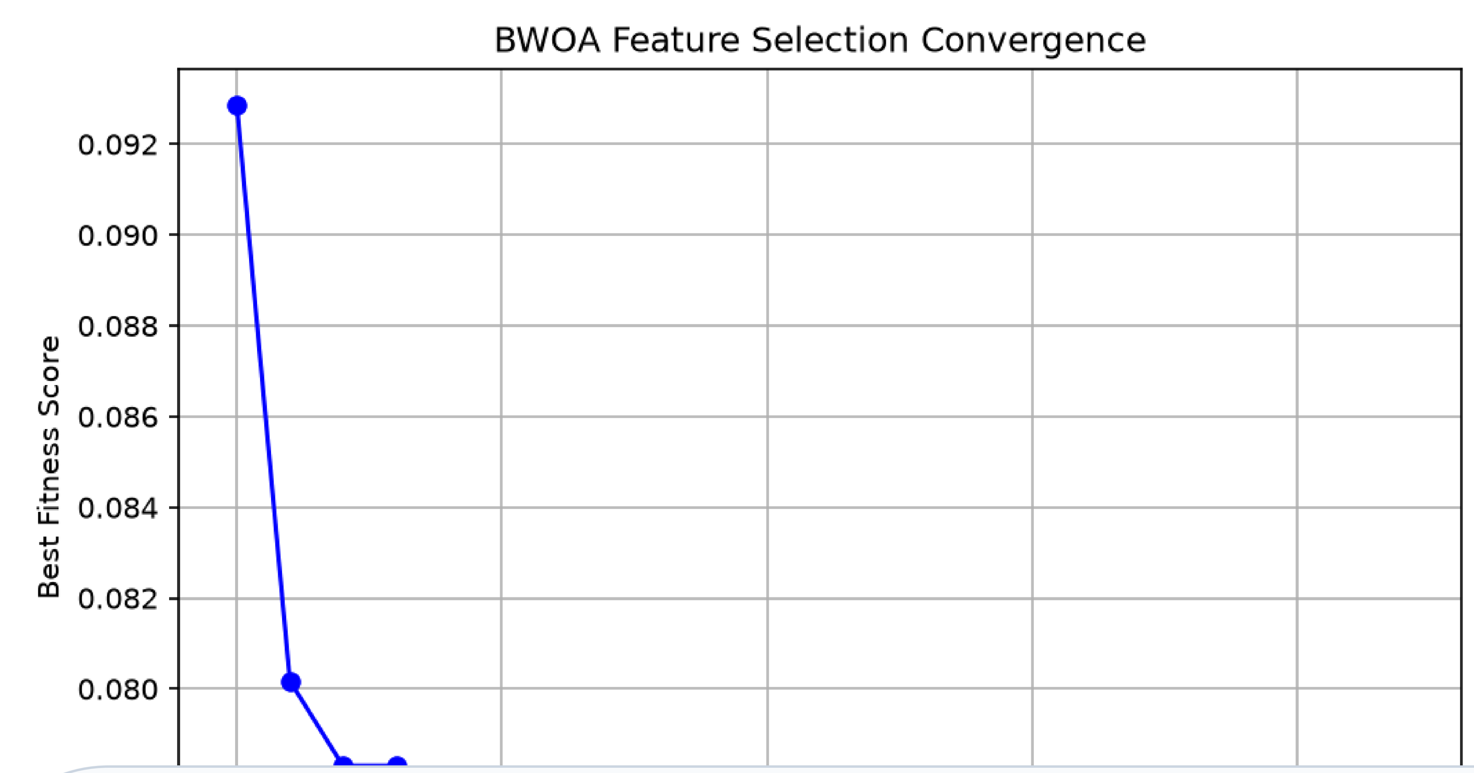
- **Data Ingestion Layer:** Promiscuous packet capture using @mhiskall282/unesco-mine-sec-cli parsing IP/TCP/Modbus packet headers at wire speed without packet drops.
- **1. Shrinking Encircling Phase:** D = |C * X*(t) - X(t)|, X(t+1) = X*(t) - A * D
where A = 2a*r1 - a, C = 2*r2, and 'a' linearly decreases from 2 to 0 over iterations.
- **2. Spiral Bubble-Net Foraging:** X(t+1) = D' * exp(b*l) * cos(2*pi*l) + X*(t)
where b=1.0, l is uniform in [-1, 1], mathematically modeling the helical hunting maneuver.
- **3. V-Shaped Binary Transfer Function:** V(x) = |x / sqrt(1 + x^2)|, X_d(t+1) = 1 - X_d(t) if rand() < V(x_d) else X_d(t)
mapping continuous velocities to discrete stochastic bit-flips without boundary saturation.
- **4. Accuracy Floor Fitness Function:** Fit(X) = alpha*(1 - Acc(X)) + (1-alpha)*(|X|/D) + Penalty(X)
where alpha=0.3 (70% weight on accuracy) and Penalty=1.0 if Acc < 0.75 or |X| < 10.
- **Spatial-Temporal CNN-LSTM Classifier:** 1D Convolutional layer (64 filters, kernel size 3) + BatchNorm + SpatialDropout(0.3) + LSTM (256 units) + Dense(64) + Softmax(5 classes).
- **Float16 Edge Quantization:** Post-training Float16 quantization compresses model memory footprint to 0.82 MB (83.2% reduction) for sub-millisecond execution on 1GB RAM edge gateways.

3. 4-Layer System Architecture

Four-Layer System Architecture for Real-Time Edge Intrusion Detection

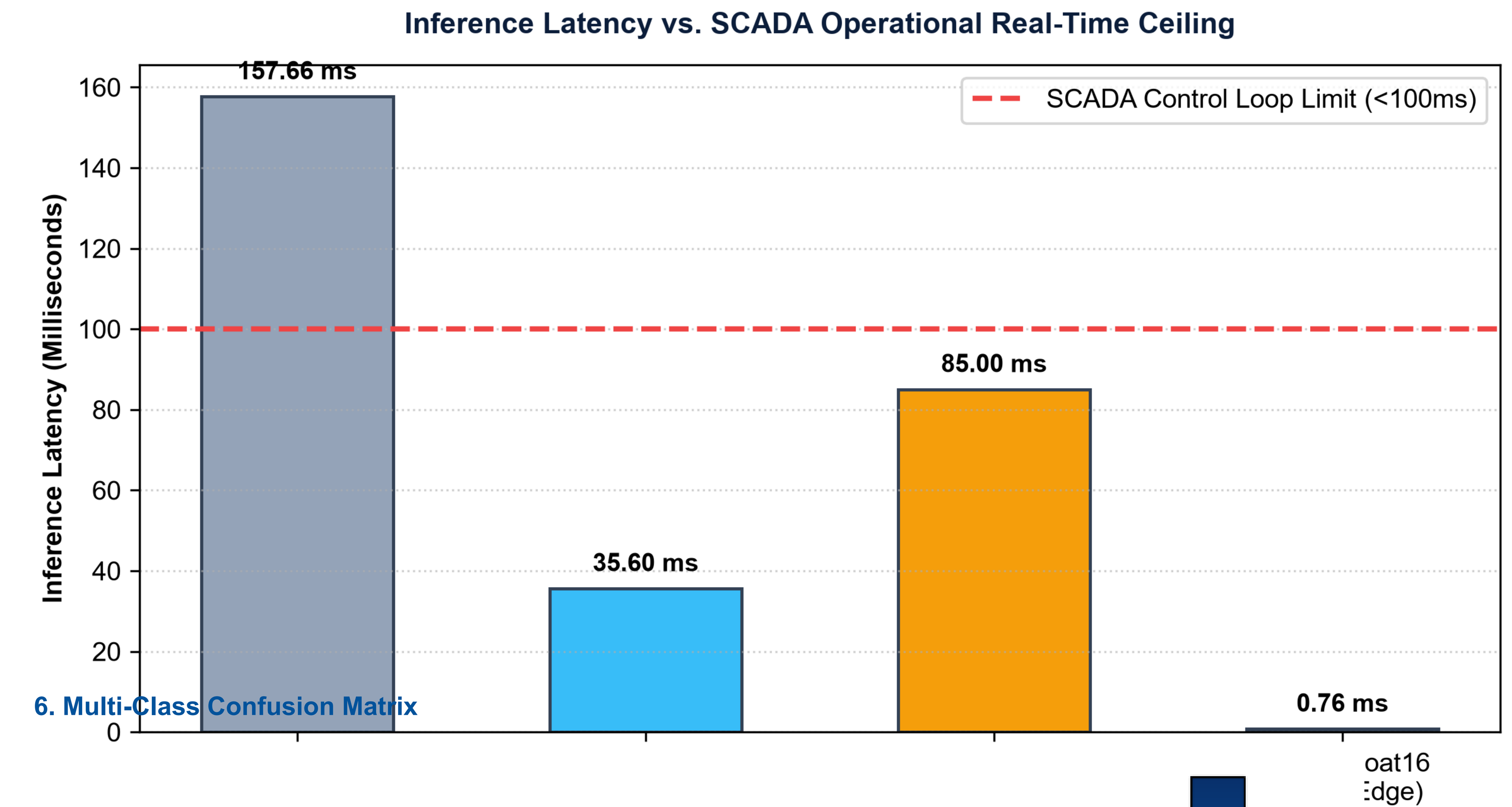


4. BWOA Optimization Analysis

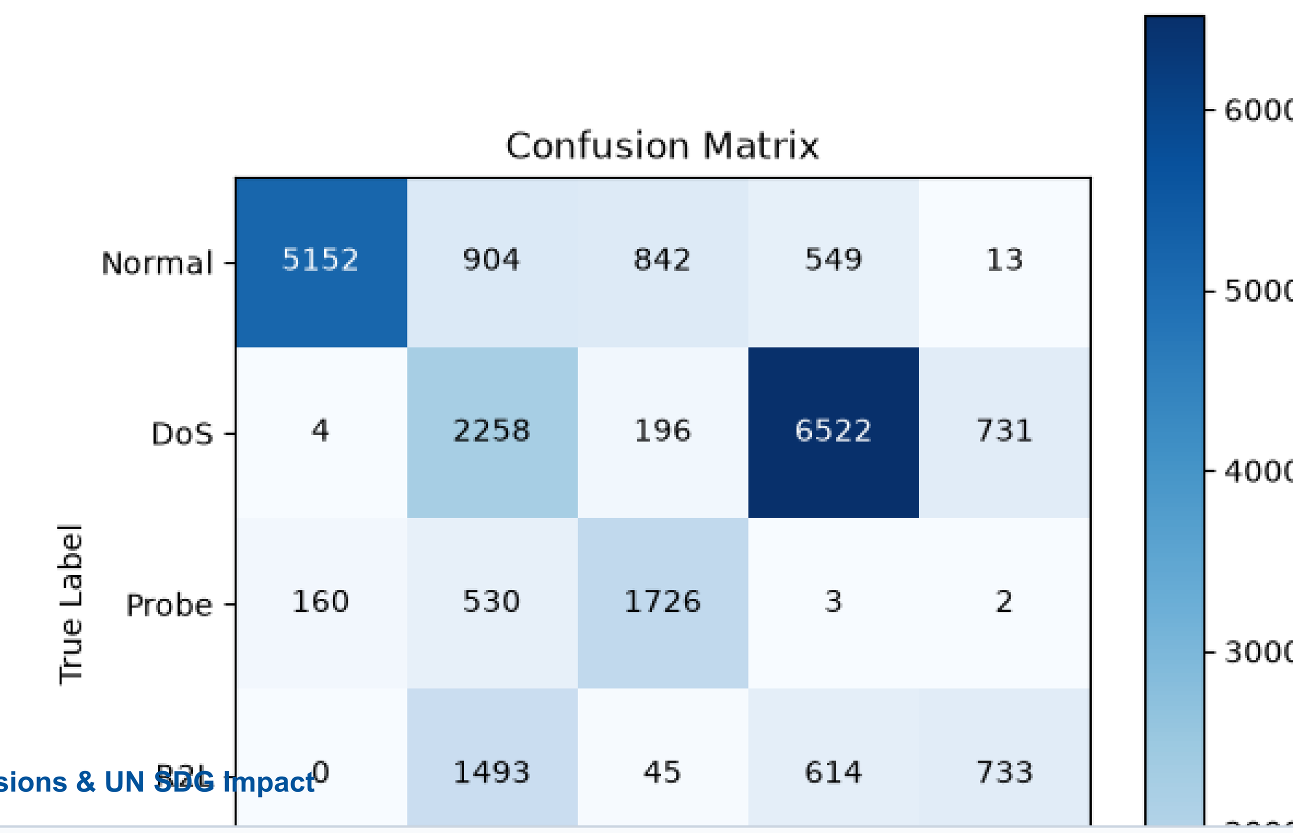


- ★ **Optimal Feature Convergence:** BWOA reached minimal fitness at iteration 23/100, maintaining 92.31% Random Forest 3-fold cross-validation accuracy on the selected 10-feature subset.
- ★ **75.61% Bandwidth Reduction:** Pruned 31 uninformative features from 41, reducing network telemetry transmission load over low-bandwidth satellite links in African mines.
- ★ **Selected 10 Vital Features:** src_bytes (volume DoS), service & flag (SCADA connection state), serror_rate & diff_srv_rate (reconnaissance), hot & su_attempted (privilege escalation).
- ★ **Transfer Learning Adaptability:** Evaluated on the 51-sensor SWaT SCADA benchmark achieving 0.12ms inference latency and 0.8650 AUC-ROC, validating cross-domain industrial suitability.

5. Empirical Hardware Latency Profile



6. Multi-Class Confusion Matrix



7. Conclusions & UN SDG Impact

- ✓ **0.76ms Edge Real-Time:** Executes single-sample inference in 0.76 ms on Raspberry Pi 4B (1GB RAM) - 207x faster than baseline, easily passing the sub-100ms SCADA limit.
- ✓ **High Detection Fidelity:** 70.56% multi-class accuracy, 96.89% precision on benign telemetry (zero false plant shutdowns), 89.04% recall on DoS attacks.
- ✓ **High Industrial ROI:** 200x-300x return averting $50k-$500k/hr outages while protecting miner lives against ventilation tampering (SDG 8 & 9).
- ✓ **Open-Source Ecosystem:** NPM sniffer package '@mhiskall282/unesco-mine-sec-cli' and 75/75 automated unit test suites ensure complete reproducibility.

8. References

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4. Mirjalili, S., & Lewis, A. (2016). Whale optimization algorithm. Advances in Eng. Software, 95, 51-67.
5. Minerals Commission of Ghana. (2024). Digital telemetry & cybersecurity compliance policy guidelines.